

Vladimir Braginsky and Kip Thorne, in Pasadena, California, October 1984. [courtesy Valentin N. Rudenko.]

Braginsky had foresight that nobody else possessed. During our 1969 meeting, and then again in 1971 and 1972, he warned me that the bars being used to search for gravitational waves have a fundamental, ultimate limitation. That limitation, he told me, comes from the laws of quantum mechanics. Although we normally think of quantum mechanics as governing tiny objects such as electrons, atoms, and molecules, if we make sufficiently precise measurements on the vibrations of a one-ton bar, we should see those vibrations also behave quantum mechanically, and their quantum mechanical behavior will ultimately cause problems for gravitational-wave detection. Braginsky had convinced himself of this by calculating the ultimate performance of Weber's piezoelectric crystals and of several other kinds of sensors that one might use to measure a bar's vibrations.

I didn't understand what Braginsky was talking about; I didn't understand his reasoning, I didn't understand his conclusion, I didn't understand its importance, and I didn't pay much attention. Other things he was teaching me seemed much more important: From him I was learning how to think about experiments, how to design experimental apparatus,